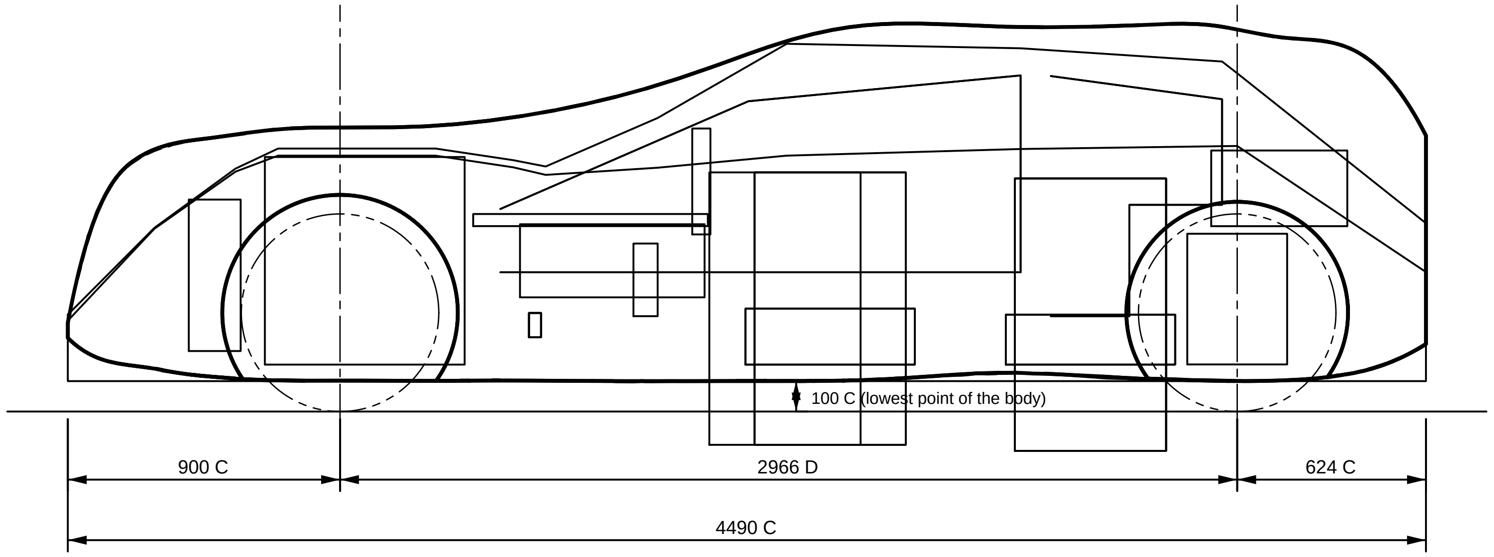
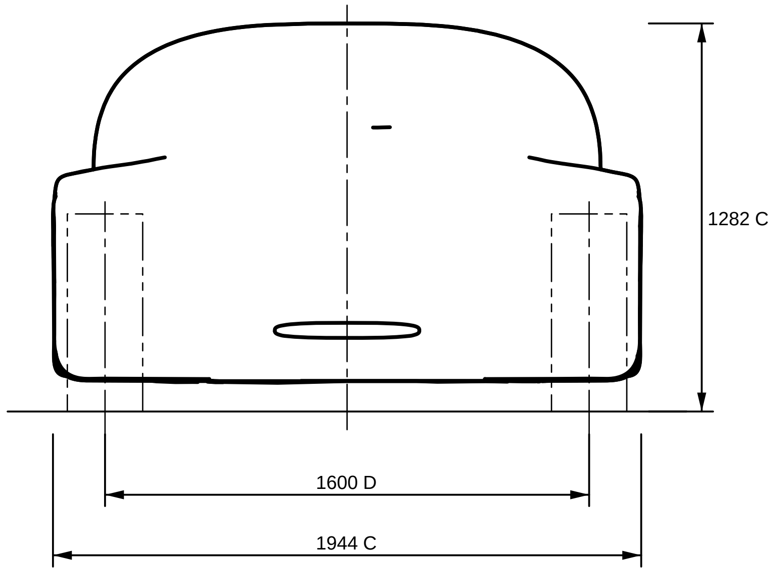
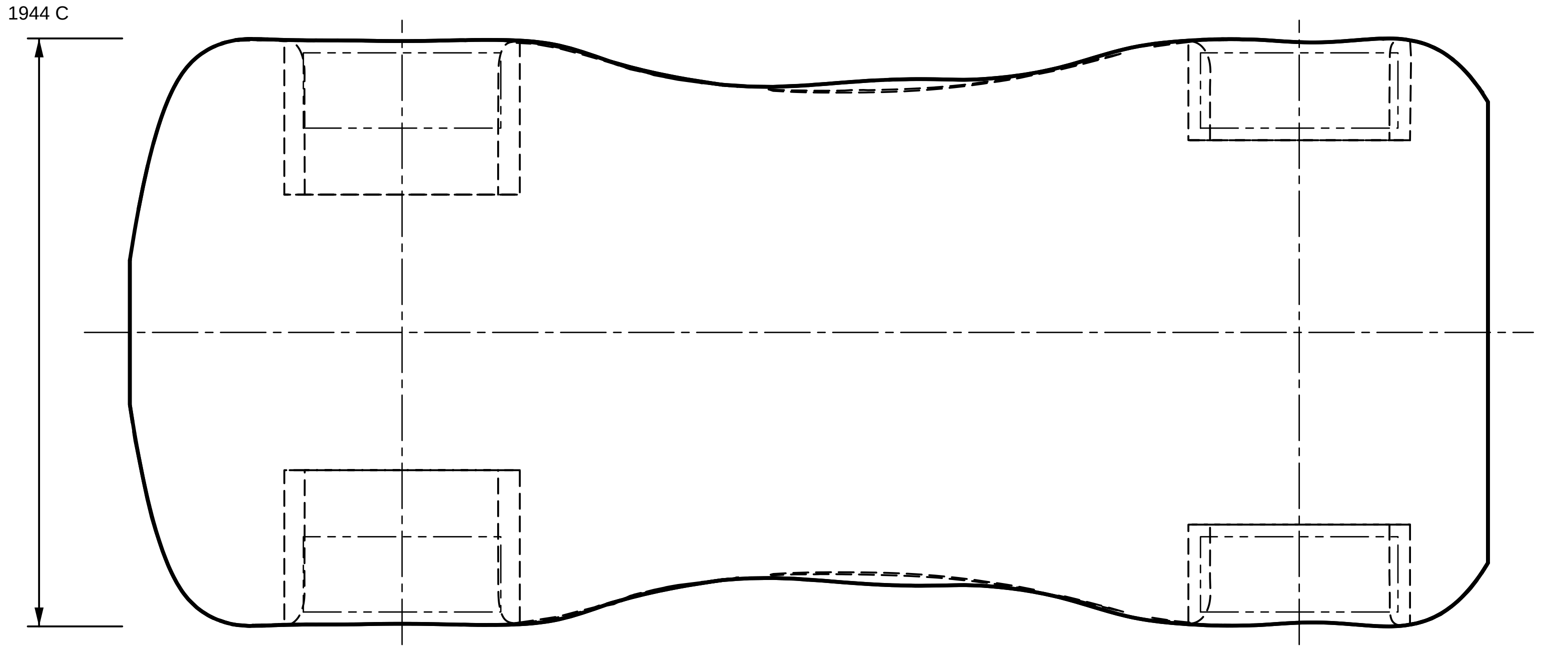
SIDE ELEVATION (left side)



FRONT VIEW



PLAN (from above; the car's right side at the top)



NOTES

Dimensions in mm. C = computed from the CAD body (kernel receipt), D = layout design choice or derivation, S = sourced (cited). Values and bases: sheet 2.

Hidden-line projection of the B-rep body (OpenCascade polygonal HLR); edges within the tessellation deflection (2 mm). Tyres and axles: phantom / centre lines.

Cutaway lines (thin) are the assembly projection: occupants, openings, controls and powertrain envelopes.

Revision = hash of the model this sheet was generated from. Any change to the body, layout or BOM gives a new revision; a sheet whose revision differs from the model's current one is superseded.

SCREENING DRAWING: generated from a software model; not checked, not approved, not for manufacture.

All masses are mass.part results (state: sourced / estimated / computed / placeholder); occupants (payload) are not parts and are not lis...

GENERAL ARRANGEMENT - VEH (ConKay library car)

Drawing no. CK-GA-VEH Sheet 1 of 3	Revision
Scale 1:25 (A3) Units mm First-angle projection	**R-52B30A9B**
Model: BODY_SHELL + VEH layout + BOM (56 items)	model hash 52b30a9b24cedaf6e465251425034704
SCREENING - NOT FOR MANUFACTURE Generated by ConKay drawing.ga 1.0.0 from cad.body (OCCT 7.7.2.0); revision = model hash	8315dd0b2b2e5f74649b3a4f3a12e6dc

BILL OF MATERIALS (component library + designed parts)

Item	Qty	Nodes	Description	Library id / material	Unit mass kg	Mass state	Source
1	1	BODY_SHELL	BODY_SHELL (designed part)	material: cfrp-hexply-8552-agp193pw	120.52	computed	cfrp-hexply-8552-agp193pw (density: Hexcel HexPly 8552 product...
2	1	BRAKES_FRONT	Wilwood AERO6 Big Brake front brake kit 140-13886, black calipers, 14...	brakes.wilwood.aero6.140-13886	23.86	sourced	www.wilwood.com/BrakeKits/BrakeKitsProdFront
3	1	BRAKES_REAR	Wilwood AERO4-MC4 Big Brake rear parking brake kit 140-13888, black c...	brakes.wilwood.aero4-mc4.140-13888	20.32	sourced	www.wilwood.com/BrakeKits/BrakeKitsProdRear
4	1	CONTROL_PACK	Ford Performance Gen 4X Coyote control pack, manual transmission M-60...	wiring.ford.control-pack.m-6017-m50hm	14.06	estimated	performanceparts.ford.com/part/M-6017-M50HM
5	1	DIFFERENTIAL	Ford Performance Mustang Super 8.8-inch IRS loaded differential housi...	differential.ford.super-8.8-irs.m-4001-88355b	71.89	estimated	www.mustang6g.com/forums/threads/the-what-things-weigh-th
6	1	ENGINE	Ford Performance 5.0L Gen 4X Mustang crate engine M-6007-M50H (10R80 ...	engine.ford.coyote-gen4x.m-6007-m50h	201.85	sourced	www.ford.com/product/50l-gen-4x-mustang-crate-engine
7	1	EXHAUST	Ford (OEM) / Bassani Exhaust after the crate engine's right-hand mani...	exhaust.s550-gt.oem-lh-manifold-midpipe.bassani-xpipe-catback	36.02	sourced	www.mustang6g.com/forums/threads/the-what-things-weigh-th
8	1	FUEL_CELL	ATL (Aero Tec Laboratories) Saver Cell, 60 L (15 US gal) SA-AA-070 (l...	fuel.atl.saver-cell.sa-aa-070	8.10	sourced	atlinside.com/Content/Downloads/ATL-Saver-Cell-Data-S
9	1	GLAZING	GLAZING (designed part)	material: glass-soda-lime	22.00	computed	glass-soda-lime (density: typical annealed float glass; streng...
10	1	RADIATOR	Cold Case Radiators Coyote swap aluminium radiator kit with fans and ...	cooling.coldcase.lmm570-5k	16.78	estimated	wheelfront.com/shop/product/cold-case-radiators-79-93-
11	2	SEAT_1, SEAT_2	RECARO Sportster GT reclining sport seat, metal seat base with reinfo...	seat.recaro.sportster-gt	15.00	sourced	www.recaro-automotive.com/fileadmin/01-products/documents/hotshee
12	2	SEAT_3, SEAT_4	Ford (OEM) S550 Mustang rear seat 2015 Mustang OEM rear seat backs + ...	seat.ford.s550-rear.oem	9.48	sourced	www.mustang6g.com/forums/threads/the-what-things-weigh-th
13	1	STEERING_RACK	Ford (OEM) S550 Mustang electric power-assisted steering (EPAS) rack ...	steering.ford.s550-epas	18.35	sourced	www.mustang6g.com/forums/threads/the-what-things-weigh-th
14	1	SUSPENSION_FRONT	Ford (OEM) S550 Mustang front suspension 2015 Mustang GT (non-Perform...	suspension.ford.s550-front.oem	78.06	sourced	www.mustang6g.com/forums/threads/the-what-things-weigh-th
15	1	SUSPENSION_REAR	Ford (OEM) S550 Mustang independent rear suspension 2015 Mustang GT O...	suspension.ford.s550-irs-rear.oem	61.83	sourced	www.mustang6g.com/forums/threads/the-what-things-weigh-th
16	4	TIRE_FL, TIRE_FR, TIRE_RL, TIRE_RR	Michelin Pilot Sport 4S 245/40ZR18 (97Y) XL, MSPN 37666	tyre.michelin.pilot-sport-4s.245-40zr18-97y-xl	11.52	sourced	wheelwiz.ca/products/m-37666
17	1	TRANSMISSION	TREMEC TKX 5-speed RWD manual TCET18085 (Ford, 26-spline input, 3.27 ...	transmission.tremec.tkx.tcet18085	50.00	estimated	tremec.com/wp-content/uploads/2023/05/TKX-Service-
18	1	TUB_A_PILLAR_L	A-pillar L (designed part)	material: aluminum-6061-t6	3.22	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
19	1	TUB_A_PILLAR_R	A-pillar R (designed part)	material: aluminum-6061-t6	3.22	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
20	1	TUB_B_POST_L	B-post L (designed part)	material: aluminum-6061-t6	3.15	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
21	1	TUB_B_POST_R	B-post R (designed part)	material: aluminum-6061-t6	3.15	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
22	1	TUB_C_POST_L	C-post L (designed part)	material: aluminum-6061-t6	2.19	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
23	1	TUB_C_POST_R	C-post R (designed part)	material: aluminum-6061-t6	2.19	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
24	1	TUB_FIREWALL_L	Firewall sheet L (designed part)	material: aluminum-6061-t6	1.48	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
25	1	TUB_FIREWALL_R	Firewall sheet R (designed part)	material: aluminum-6061-t6	1.48	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
26	1	TUB_FLOOR_L	Floor sheet L (designed part)	material: aluminum-6061-t6	5.40	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
27	1	TUB_FLOOR_R	Floor sheet R (designed part)	material: aluminum-6061-t6	5.40	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
28	1	TUB_FW_CPOST	Firewall centre post (designed part)	material: aluminum-6061-t6	0.55	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
29	1	TUB_FW_LOWER	Firewall lower cross (designed part)	material: aluminum-6061-t6	17.45	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
30	1	TUB_FW_POST_L	Firewall post L (designed part)	material: aluminum-6061-t6	0.73	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
31	1	TUB_FW_POST_R	Firewall post R (designed part)	material: aluminum-6061-t6	0.73	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
32	1	TUB_HEEL_CROSS	Heel cross (designed part)	material: aluminum-6061-t6	3.87	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
33	1	TUB_KICK_L	Sill kick L (designed part)	material: aluminum-6061-t6	4.80	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
34	1	TUB_KICK_R	Sill kick R (designed part)	material: aluminum-6061-t6	4.80	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
35	1	TUB_QTR_L	Rear quarter box L (designed part)	material: aluminum-6061-t6	7.97	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
36	1	TUB_QTR_R	Rear quarter box R (designed part)	material: aluminum-6061-t6	7.97	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
37	1	TUB_RAIL_FL	Front rail L (designed part)	material: aluminum-6061-t6	4.60	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
38	1	TUB_RAIL_FR	Front rail R (designed part)	material: aluminum-6061-t6	4.60	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
39	1	TUB_REAR_BULKHEAD	Rear bulkhead (designed part)	material: aluminum-6061-t6	5.52	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
40	1	TUB_REAR_CROSS	Rear end cross (designed part)	material: aluminum-6061-t6	4.84	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
41	1	TUB_REAR_FLOOR_L	Rear floor sheet L (designed part)	material: aluminum-6061-t6	1.66	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
42	1	TUB_REAR_FLOOR_R	Rear floor sheet R (designed part)	material: aluminum-6061-t6	1.66	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
43	1	TUB_ROOF_BOW	Roof bow (B-hoop top) (designed part)	material: aluminum-6061-t6	3.56	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
44	1	TUB_ROOF_LINER	Roof liner (bonded to the skin) (designed part)	material: aluminum-6061-t6	4.00	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
45	1	TUB_ROOF_RAIL_L	Roof rail L (designed part)	material: aluminum-6061-t6	3.27	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
46	1	TUB_ROOF_RAIL_R	Roof rail R (designed part)	material: aluminum-6061-t6	3.27	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
47	1	TUB_ROOF_RAIL_R_L	Rear roof rail L (designed part)	material: aluminum-6061-t6	2.32	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
48	1	TUB_ROOF_RAIL_R_R	Rear roof rail R (designed part)	material: aluminum-6061-t6	2.32	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
49	1	TUB_SCUTTLE	Scuttle cross (designed part)	material: aluminum-6061-t6	4.20	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
50	1	TUB_SEAT_CROSS	Seat cross (designed part)	material: aluminum-6061-t6	3.01	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
51	1	TUB_SILL_L	Sill L (designed part)	material: aluminum-6061-t6	11.57	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
52	1	TUB_SILL_R	Sill R (designed part)	material: aluminum-6061-t6	11.57	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
53	1	TUB_TUNNEL_F	Tunnel, gearbox cell (designed part)	material: aluminum-6061-t6	4.41	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
54	1	TUB_TUNNEL_R	Tunnel, driveshaft cell (designed part)	material: aluminum-6061-t6	5.62	computed	aluminum-6061-t6 (density: engineering material library (lib/a...

GENERAL ARRANGEMENT - VEH (ConKay library car)

Drawing no. CK-GA-VEH   Sheet 2 of 3

Scale 1:25 (A3)   Units mm   First-angle projection

Model: BODY_SHELL + VEH layout + BOM (56 items)

SCREENING - NOT FOR MANUFACTURE

Generated by ConKay drawing.ga 1.0.0 from cad.body (OCCT 7.7.2.0); revision = model hash

Revision

R-52B30A9B

model hash 52b30a9b24cedaf6e465251425034704
8315dd0b2b2e5f74649b3a4f3a12e6dc

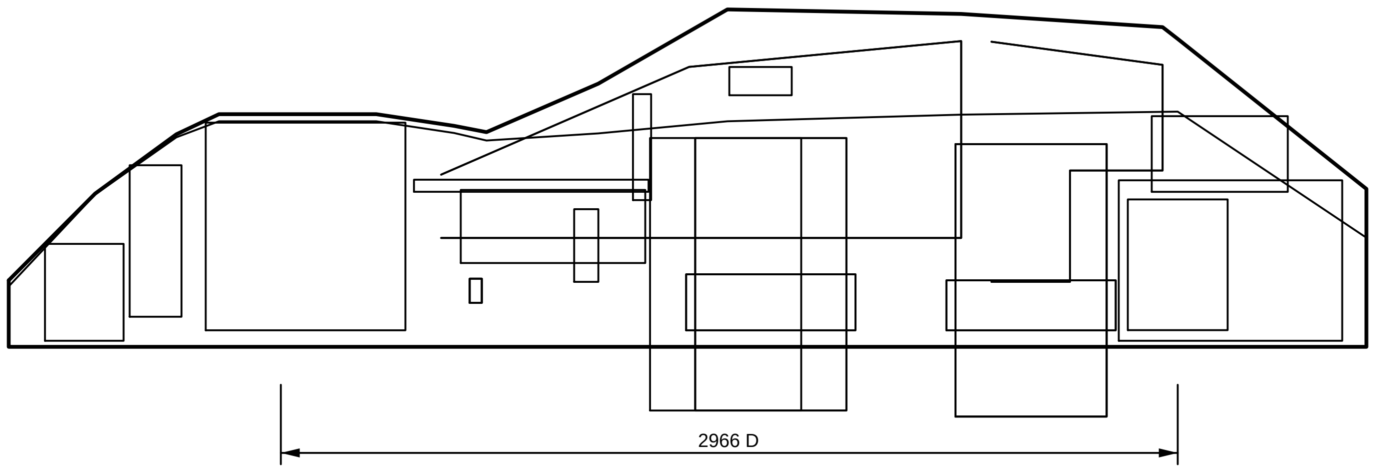
PACKAGING DIMENSIONS AND FIT

H30 185 mm assumed SAE J1100 (2001)
L53 820 mm calculated SAE J1100 (2001)
L40 25 deg assumed SAE J1100 (2001)
H61-roof 928 mm calculated SAE J1100 (2001)
BOF 203 mm sourced SAE J1100 (2001)
W9 350 mm assumed SAE J1100 (2001)
H17 605 mm assumed SAE J1100 (2001)
eyellipse-X 206.4 mm sourced SAE J941 (2010) Table 2
eyellipse-centroid-z 978 mm sourced SAE J941 (2010) Eq. 4
belt 845 mm calculated ANSUR II acromion, sitting

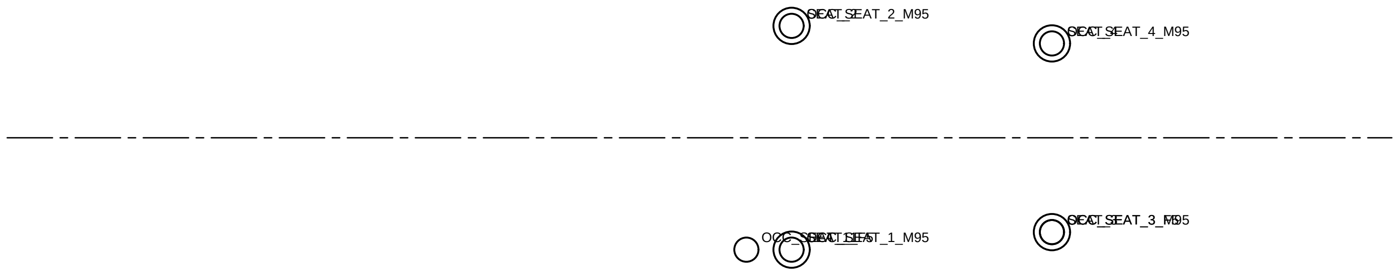
FIT CHECKS

PASS head-M95 M95 head top at 1139 mm (x 2993 mm), roof at that station 1219 mm, inner gap 76 mm
PASS head-F5 F5 head top at 997 mm (x 2777 mm), roof at that station 1238 mm, inner gap 237 mm
PASS head-M95-rear rear M95 head top at 1119 mm (x 3854 mm), roof at that station 1174 mm, inner gap 52 mm
PASS heel-floor driver heel z 155 mm, floor 155 mm
PASS rear-leg M95 rear occupant reaches the floor in the vertical-shin posture
PASS four-seats 4 seat positions placed
PASS powertrain-cabin engine rear face 1412 mm, firewall 1462 mm, gap 50 mm. hood raised 191 mm for the library radiator; shoulder raised 70 mm...
PASS powertrain-layout The design's powertrain is a front engine (gasoline): the compiler's engine, gearbox and cooling are drawn where the pac...
PASS wheel-footwell front tyre swept +/-39.8 deg vs the driver foot, gap 278 mm (separating-axis lower bound)
PASS ingress skin door aperture (roof rail) 1246 mm, M95 head top 1139 mm, gap 107 mm. Tub outline top is 1110 mm; the skin cut is the roof rai...
PASS containment every non-exterior bounding box is inside the body envelope
PASS b-pillar B-pillar station 3250 mm, front H 2620 mm, rear H 3481 mm
PASS cooling-power radiator stays the library core; required power at the brief speed is 118815 W (coolant-loop is a separate solver)
WARN spare no spare tire is specified, so none is placed
WARN design-quality 45 styling controls. dash-to-axle 23% of the wheelbase, hood 184 mm vs the arch top, rake 60.0 deg, greenhouse/body 0.50, t...

SIDE CUTAWAY (assembly projection, left side)



PLAN - SEAT H-POINTS (assembly)



Projected from the assembly. Envelopes are not kernel solids.

Packaging and occupant layout

Drawing no. CK-GA-VEH   Sheet 3 of 3
Scale 1:25 (A3)   Units mm   First-angle projection
Model: VEH assembly

SCREENING - NOT FOR MANUFACTURE
Projected from assembly.vehicle; revision = model hash

Revision
R-52B30A9B
model hash 52b30a9b24cedaf6e465251425034704
8315dd0b2b2e5f74649b3a4f3a12e6dc